THE ZERO SCHEME OF A REGULAR SECTION OF AN AMPLE INVERTIBLE SHEAF ON A CONNECTED PROPER SCHEME OF DEPTH AT LEAST TWO IS CONNECTED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Vanishing of H^0 and H^1 of negative twists of an ample invertible sheaf. Stacks Project, Varieties, Lemmas lemma-vanishin-h0-negative (tag 0FD7) and lemma-vanishin-h1-negative (tag 0FD8). The two vanishing statements contradicted by the disconnection of the zero scheme.
- Regular sections and their zero schemes as effective Cartier divisors. Stacks Project, Divisors, Definition divisors-definition-regular-section (tag 0AYQ). Guarantees that the powers s^n are regular sections, which is what makes the comparison of zero schemes valid.

Source reference. Original reference 01WY: sources/stacks-chow/tags.txt, source lines 2499–2499. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BUG: sources/stacks-chow/tags.txt, source lines 14874–14874. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FD7: sources/stacks-chow/tags.txt, source lines 19207–19207. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FD8: sources/stacks-chow/tags.txt, source lines 19208–19208. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k. Let $\mathcal{L}$ be an ample invertible $\mathcal{O}_X$-module. Let $s \in \Gamma(X, \mathcal{L})$. Assume*

(1) *s is a regular section (Divisors, Definition 01WY),*
(2) *for every closed point $x \in X$ we have depth$(\mathcal{O}_{X,x}) \geq 2$, and*
(3) *X is connected.*

Then the zero scheme $Z(s)$ of s is connected.

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Proof. Since s is a regular section, so is $s^n \in \Gamma(X, \mathcal{L}^{\otimes n})$ for all $n > 1$. Moreover, the inclusion morphism $Z(s) \to Z(s^n)$ is a bijection on underlying topological spaces. Hence if $Z(s)$ is disconnected, so is $Z(s^n)$. Now consider the canonical short exact sequence

$$0 \to \mathcal{L}^{\otimes -n} \xrightarrow{s^n} \mathcal{O}_X \to \mathcal{O}_{Z(s^n)} \to 0$$

Consider the k-algebra $R_n = \Gamma(X, \mathcal{O}_{Z(s^n)})$. If $Z(s)$ is disconnected, i.e., $Z(s^n)$ is disconnected, then either R_n is zero in case $Z(s^n) = \emptyset$ or R_n contains a nontrivial idempotent in case $Z(s^n) = U \amalg V$ with $U, V \subset Z(s^n)$ open and nonempty (the reader may wish to consult Lemma 0BUG). Thus the map $\Gamma(X, \mathcal{O}_X) \to R_n$ cannot be an isomorphism. It follows that either $H^0(X, \mathcal{L}^{\otimes -n})$ or $H^1(X, \mathcal{L}^{\otimes -n})$ is nonzero for infinitely many positive n. This contradicts Lemma 0FD7 or 0FD8 and the proof is complete. □